DEGREE: MSc Data Analytics

Module: Computer Vision and Artificial Intelligence

Assignment Title: AI-Driven Image Recognition for Automated Plant Disease Diagnosis in Modern Smart Farming

Assignment Type: Practical skills assessment (including production of an artefact)

Word Limit: 3000 words (+/- 300)

Weighting: 50%

Issue Date: 02-12-2025

Submission Date: 06-02-2026

Feedback Date: 10-03-2026

Issued by: Dr. Fahmid Al Farid

Plagiarism:

When submitting work for assessment, students should be aware of the InterActive/Canvas guidance and regulations in concerning plagiarism. All submissions should be your own, original work.

You must submit an electronic copy of your work. Your submission will be electronically checked.

Learner declaration
I certify that the work submitted for this assignment is my own and research sources are fully acknowledged. **Student signature:** **Date:**

Harvard Referencing:

The Harvard Referencing System must be used. The Wikipedia, UKEssays.com or similar websites must **not** be used or referenced in your work.

Introduction

Learning Outcomes:

LO1. Demonstrate the understanding of various techniques for working with images and computer vision using artificial intelligence.

LO2. Use deep learning and artificial intelligence to write algorithms and use models for image processing.

LO3. Implement Python code for applying computer vision and artificial intelligence for applications like face recognition, feature detection and matching, motion estimation, motion tracking, image classification, and object recognition.

Assessment Criteria: Weighting 50%
3000 words

Tasks:

(the task is related to **LO1**, **LO2**, and **LO3**). Please research and find deep learning architectures which are designed for the purpose of image classification. You need to practically implement them regarding the number of learning parameters and their scalability. You are also going to itemise the advantages and disadvantages of any of them.

Analysis Scenario:

Imagine you are working as a machine learning specialist alongside a research team focused on plant disease analysis within the agricultural sector. The team's goal is to design and deploy an image-based classification model capable of accurately identifying different crop diseases from leaf images. Farmers and agricultural experts face ongoing difficulties in detecting and managing plant infections at an early stage, and a reliable automated detection system would greatly support timely treatment and crop protection.

Your task is to implement a **distinct deep learning image classification method** using a framework such as **TensorFlow** or **PyTorch**, applied to a real-world dataset containing images of multiple crops exhibiting various diseases. Each image in the dataset is annotated by disease class, and your model should learn to recognize and classify these plant disease categories effectively.

Task Requirements:

You must use an agricultural plant disease image dataset of your choice from a freely available online source. Clearly mention the exact source (URL or repository location where the dataset was obtained.

When presenting your implementation, include screenshots (not copy-pasted code) of your Python code and the resulting classification outputs (e.g., prediction labels, confusion matrix, or visualizations). This is required to verify that you have implemented the solution yourself.

Abstract

- Write a brief overview of your project, summarizing the objective (e.g., automated crop disease detection), dataset used, the deep learning method applied, and the key results achieved. This should help the reader quickly understand the scope and value of your work.

1. Introduction

   - Introduce the challenges in agricultural plant disease management and explain why early detection is important. Describe how deep learning and image classification can provide an effective solution. State your objective clearly — developing a model for automated plant disease diagnosis.

2. Literature Review

   - Summarize existing approaches used for plant disease detection via computer vision. Discuss a few key models (e.g., CNNs, transfer learning frameworks) and their outcomes. Mention gaps such as lack of scalability, accuracy, or speed that your approach aims to improve or build upon.

3. Problem Formulation

   - Define the practical issue: plant disease spreads fast and is hard to monitor manually across large fields. Stress the importance of a reliable, automated system. Clarify the goals of your project: building a model that classifies plant leaf images by disease type.

4. Dataset Preparation

   - Describe the dataset used (e.g., PlantVillage, Kaggle). Include the number of classes, images, and how data is labeled. Explain how you prepared it — resizing, augmentation, normalization, and train-test splits. This ensures reproducibility and clarity in your work.

5. Model Architecture

   - Describe the deep learning model used — whether a custom CNN or a pre-trained model like ResNet or MobileNet. Justify your selection based on model performance, size, training time, and relevance to the task. Include a short summary of layers/components.

6. Implementation

- Summarize how your code is structured using Python and a deep learning framework (e.g., TensorFlow or PyTorch). Explain each core component: data pipeline, model instantiation, training loop, and evaluation. Use brief code snippets only if needed to clarify logic.

7. Model Evaluation

    - Present the model's performance using relevant metrics such as accuracy, precision, recall, F1-score, and confusion matrix. Discuss how well the model performed on unseen data and where it might struggle (e.g., class imbalance, similar-looking symptoms).

8. Conclusion

    - Briefly summarize your model's impact and what it demonstrates. Discuss any limitations, such as overfitting or generalizability to other crops. Suggest improvements more diverse datasets, real-time deployment, or integration with field-based monitoring tools.

    - Emphasize the importance of clarity and thorough explanation in each section to ensure a comprehensive understanding of the entire process, from problem formulation to evaluation.

Submission Instructions:

- Create a practical analysis within the designated word limit (2000 - 3000 words).
- Integrate Python code snippets, visualizations, and pertinent documentation seamlessly into the analysis.
- Maintain clarity, organization, and logical flow in presenting your result analysis.
- Utilize the BSBI assignment template provided on Canvas for document preparation.
- Employ the Harvard referencing style for your bibliography.
- Refer to the Essay-Guide on Canvas for additional instructions.
- Submit your assignment electronically by the stipulated deadline.

Students are encouraged to explore additional reputable sources in the digital library beyond the list provided to strengthen their individual research and critical analysis skills. The unit lecturer can help with recommending additional literature.

The criteria for successfully passing the units with two assessment components require passing both components successfully. Passing marks for BSc and MA/MSc levels are different and are stated at the end of the assignment brief. Please find the examples below:

· **Example 1** (e.g., pass mark: 50)
· Component 1 (50% weighting) - 55
· Component 2 (50% weighting) - 55
· Final average: 55
· Result: **Pass (both components passed and therefore the unit is passed)**

· **Example 2** (e.g., pass mark: 50)
· Component 1 (50% weighting) – 80
· Component 2 (50% weighting) – 30
· Final average: 55
· Result: **Fail (because although the average is above 50, one of the components is a failure**)

GRADING DESCRIPTORS: LEVEL 7

EXPERIMENTATION & INNOVATION								
	FAIL			PASS				
Threshold Criteria	**0-29%**	**30-39%**	**40-49%**	**50-59%**	**60-69%**	**70-79%**	**80-89%**	**90-100%**
Deals with complex issues both systematically and creatively demonstrating self-direction and originality in tackling and solving problems	Little to no ability to use techniques to deal with complex issues systematically (including those of ethics and sustainability) and creatively to solve problems and/or make decisions.	Low utilisation of established techniques to deal with complex issues systematically (including those of ethics and sustainability) and creatively to solve problems and/or make decisions, but with limitations in techniques or approach.	Limited research or advanced scholarship to their area of study by using a range of information and established and advanced techniques	Competent understanding of solving problems, through own research or advanced scholarship displaying a comprehensive understanding of established and advanced techniques	Good understanding of solving problems through own research and advanced scholarship critically selecting and displaying a comprehensive understanding of established and advanced techniques.	Very Good problem-solving skills displaying a comprehensive understanding of techniques applicable to their own research or advanced scholarship	Excellent range of extremely well-developed problem-solving displaying an understanding of techniques applicable to their own research or advanced scholarship beyond which is taught.	Exceptional problem-solving skills with sophisticated evaluation and application of a wide range of advanced information and techniques to undertake projects.
Comprehensive understanding of techniques applicable to their own research or advanced scholarship	Little to no understanding of techniques applicable to their own research or advanced scholarship or their limitations and ambiguities.	Low understanding of techniques applicable to their own research or advanced scholarship including their limitations and ambiguities.	Limited understanding of key techniques applicable to their own research or advanced scholarship including their limitations and ambiguities.	Competent understanding of techniques applicable to their own research or advanced scholarship including their limitations and ambiguities	Good understanding of techniques applicable to their own research or advanced scholarship and a some understanding of more specialised techniques.	Very good understanding of techniques applicable to their own research or advanced scholarship and a some understanding of more specialised techniques.	Excellent understanding of techniques applicable to their own research or advanced scholarship and mastery of some more specialised areas.	Exceptional understanding of techniques applicable to their own research or advanced scholarship and mastery of some more specialised areas.

GRADING DESCRIPTORS: LEVEL 7

RESEARCH & ANALYSIS								
	FAIL			PASS				
Threshold Criteria	**0-29%**	**30-39%**	**40-49%**	**50-59%**	**60-69%**	**70-79%**	**80-89%**	**90-100%**
Systematic understanding of knowledge, and a critical awareness of current problems and/or new insights, much of which is at, or informed by, the forefront of their academic discipline, field of study or area of professional practice	Little to no knowledge of the subject with limited breadth or depth or deficiencies in major areas or currency.	Low knowledge of the subject lacking coherence, breadth, or detail with only some reference to ideas or arguments at the forefront of any part of the subject.	Limited knowledge to deal with terminology, facts and concepts some of which is informed by the forefront of defined areas of the subject.	Competent knowledge of ideas or arguments at the forefront of any part of the subject sufficient to deal with current issues in the discipline, generally more descriptive than critical or analytical.	Good knowledge of ideas or arguments at the forefront of any part of the subject showing a clear, critical insight into the discipline as whole and current issues/problems.	Very good knowledge of ideas or arguments at the forefront of the subject some of which are significantly beyond what has been taught and show a critical insight into the discipline and current issues/problems.	Excellent knowledge of ideas or arguments at the forefront of the subject many of which are significantly beyond what has been taught and show a critical insight into the discipline and current issues/problems.	Exceptional knowledge of ideas or arguments at the forefront of the subject most of which are significantly beyond what has been taught and show a critical insight into the discipline and current issues/problems.
Conceptual understanding that enables the student to display originality in the application of knowledge	Little to no conceptual understanding or argument and a focus on descriptive explanations which do not comment on arguments of others or alternative views.	Low conceptual understanding and arguments are weak or poorly constructed, and the work does not critically evaluate the arguments of others or consider alternative views.	Limited conceptual understanding and argument construction with critical evaluation of alternative views or comment on advanced scholarship.	Competent conceptual understanding and argument construction with critical evaluation of a range of views and consistent engagement with advanced scholarship.	Good conceptual understanding which critically evaluate and synthesise other views and information with a thoughtful interpretation of advanced scholarship.	Very good conceptual understanding which systematically synthesises a wide range of views with a critical insight into advanced scholarship.	Excellent conceptual understanding which critically apply a wide range of views through a perceptive use of advanced scholarship.	Exceptional conceptual understanding of publishable quality with systematic engagement and usage of advanced scholarship.

GRADING DESCRIPTORS: LEVEL 7

ENGAGING WITH PRACTICE								
	FAIL			**PASS**				
Threshold Criteria	**0-29%**	**30-39%**	**40-49%**	**50-59%**	**60-69%**	**70-79%**	**80-89%**	**90-100%**
Practical understanding of how established techniques of research and enquiry are used to create and interpret knowledge in the discipline	Little to no evidence of background investigation, analysis, research, enquiry, ethical awareness, and/or study.	Low evidence of background investigation, analysis, research, enquiry, ethical awareness, and/or study.	Limited background investigation, analysis, research, enquiry , ethical awareness, and/or study using established techniques, with the ability to extract relevant points.	Competent investigation, analysis, research, enquiry, ethical awareness, and/or study using established techniques accurately, and can critically appraise and use academic sources.	Good background investigation, analysis, research, enquiry, ethical awareness, and/or study using established techniques accurately, and possesses a well-developed ability to critically appraise a wide range of sources.	Very good, independent, extensive and appropriate investigation, analysis, research, enquiry, ethical awareness, and/or study beyond the usual range, and critically evaluates this to advance the work and/or direct arguments.	Excellent independent, extensive and appropriate investigation, analysis, research, enquiry, ethical awareness, and/or study well beyond the usual range, and critically evaluates this to advance the work and/or direct arguments.	Exceptional investigation, analysis, research, enquiry, ethical awareness, and/or study which demonstrates carefully considered depth and breadth and critically synthesises this to advance the work and/or direct arguments.
Originality in the application of knowledge	Little to no technical, creative or artistic skills related to their area of study.	Low technical, creative or artistic skills related to their area of study.	Limited technical, creative or artistic skills required for area of study.	Competent technical, creative or artistic skills required for area of study.	Good technical, creative or artistic skills required for area of study.	Very good range of technical, creative or artistic skills.	Excellent range of technical, creative or artistic skills	Exceptional range of technical, creative or artistic skills
Independently advance your own knowledge and understanding, and to develop new skills to a high level.	Little to no contribution to group activity and/or undertaking further training at a high/advanced level.	Low contribution to group activity and/or undertaking further training at a high/advanced level.	Limited contribution to group activity and/or undertaking further training at a high/advanced level.	Competent contribution to group activity and/or independently undertakes further training at a high/advanced level.	Good contribution to group activity and/or independently undertakes further training at a high/advanced level with an understanding of team roles	Very good contribution to group activity and/or independently undertakes further training at a high/advanced level with an understanding of team roles	Excellent contribution to group activity and/or independently undertakes further training at a high/advanced level with teamwork and leadership	Exceptional contribution to group activity and/or independently undertakes further training at a high/advanced level with teamwork and strong leadership.

GRADING DESCRIPTORS: LEVEL 7

REALISATION & COMMUNICATION								
	FAIL			**PASS**				
Threshold Criteria	**0-29%**	**30-39%**	**40-49%**	**50-59%**	**60-69%**	**70-79%**	**80-89%**	**90-100%**
Communicate information, ideas, problems and solutions to both specialist and non-specialist audiences.	Little to no clarity in the communication of ideas, problems and solutions to audiences.	Low clarity in the communication of ideas, problems and solutions to audiences.	Limited clarity in the communication of ideas, problems and solutions to audiences.	Competent communication of ideas, problems and solutions to audiences.	Good, confident and clear communication of ideas, problems and solutions to audiences in a range of means / media.	Very good, confident and clear communication of ideas, problems and solutions to audiences in a range of means / media.	Excellent communication of ideas, problems and solutions to audiences in a range of means / media.	Exceptional communication of ideas, problems and solutions to audiences in a range of means / media.

GRADING DESCRIPTORS: LEVEL 7

PERSONAL & PROFESSIONAL CONNECTIVITY								
	FAIL			**PASS**				
Threshold Criteria	**0-29%**	**30-39%**	**40-49%**	**50-59%**	**60-69%**	**70-79%**	**80-89%**	**90-100%**
Independently advance your own knowledge and understanding, and develop new skills to a high level.	Little to no contribution to group activity and/or undertaking further training at a high/advanced level.	Low contribution to group activity and/or undertaking further training at a high/advanced level.	Limited contribution to group activity and/or undertaking further training at a high/advanced level.	Competent contribution to group activity and/or independently undertakes further training at a high/advanced level.	Good contribution to group activity and/or independently undertakes further training at a high/advanced level with an understanding of team roles	Very good contribution to group activity and/or independently undertakes further training at a high/advanced level with an understanding of team roles	Excellent contribution to group activity and/or independently undertakes further training at a high/advanced level with teamwork and leadership	Exceptional contribution to group activity and/or independently undertakes further training at a high/advanced level with teamwork and strong leadership.
Qualities and transferable skills necessary for employment requiring: (a) the exercise of initiative, ethical and personal responsibility (b) decision-making in complex and unpredictable contexts	Little to no ability to manage learning and/or exercise initiative, ethical and personal responsibility and/or decision-making in complex and unpredictable situations	Low ability to manage learning and/or exercise initiative, ethical and personal responsibility and/or decision-making in complex and unpredictable situations	Limited ability to manage learning and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations	Competent ability to manage learning, and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations	Good ability to systematically manage learning, and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations	Very good ability to systematically manage learning, and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations.	Excellent ability to manage learning on own initiative, and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations	Exceptional ability to manage learning on own initiative, and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations
	Little to no use of appropriate terminology, limited vocabulary and many errors in spelling, grammar and syntax.	Low use of appropriate terminology, with many errors in spelling, vocabulary and syntax.	Limited expression, style and appropriate vocabulary with errors in spelling, grammar and syntax which affect understanding.	Competent expression, style, and appropriate vocabulary with some errors in spelling, grammar and syntax which do not affect understanding.	Good expression, style and appropriate vocabulary with some errors in spelling, grammar and syntax.	Very good expression, style and appropriate vocabulary with minimal errors in spelling, grammar and syntax.	Excellent expression, style and appropriate vocabulary with minimal errors in spelling, grammar and syntax.	Exceptional expression, style and appropriate vocabulary with no errors in spelling, grammar and syntax.

GRADING DESCRIPTORS: LEVEL 7

	Little to no evidence of basic numeracy or digital literacy, hardware and software skills	Low evidence of basic numeracy or digital literacy, hardware and software skills competency.	Limited evidence of numeracy or digital literacy, hardware and software skills competency.	Adequate evidence of numeracy or digital literacy, hardware and software skills competency.	Good evidence of numeracy or digital literacy, hardware and software skills competency.	Very good evidence of numeracy or digital literacy, hardware and software skills	Excellent evidence of numeracy or digital literacy, hardware and software skills competency.	Exceptional evidence of numeracy or digital literacy, hardware and software skills competency.

	competency.					competency.		
	Does not demonstrate achievement of professional competence when assessed against the requirements of a professional, statutory or regulatory body (PSRB).			The student has demonstrated achievement of professional competence when assessed against the requirements of a PSRB.				
	Inaccurate use of terminology with limited vocabulary and many errors in spelling, grammar and syntax. Inaccurate terminology, with many errors in spelling, vocabulary and syntax.			The student has adhered to the appropriate rules and/or conventions set by regulators or the industry.				